Heterogeneous firms

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Investment and firm heterogeneity

- **Workshop so far:** household heterogeneity in NK model
  - Changes transmission of monetary policy to consumption!
- **Parallel literature**: firm heterogeneity, though mostly in flexible-price models
  - **Q:** How does this change transmission of monetary policy to investment?
- Paper in preparation for Journal of Economic Literature, with Tom Winberry

Plan

1. Canonical HANK model with **representative firm** investment
2. New transmission mechanisms for I with added **firm heterogeneity**
3. Complementarities from combining heterogeneous households and firms

Adding investment to canonical HANK model

Adding investment with rep. firm

- No investment in canonical HANK model so far. Let's change this!
- Representative firm produces $Y_t = K_{t-1}^{\alpha} N_t^{\nu}$, faces capital adjustment costs, solves

$$\max_{K_t, N_t} \sum_{t \geq 0} \prod_{s \leq t} \frac{1}{1 + r_s} D_t \qquad D_t \equiv Y_t - w_t N_t - I_t - \frac{1}{2}\phi\left(\frac{I_t}{K_{t-1}} - \delta\right)^2 K_{t-1} + T_t^f$$

- Flexible prices and perfect competition: $w_t = MPN_t = \nu Y_t / N_t$
- Households take N_t as given, receive labor income $w_t N_t$ and transfers T_t^h
- Government issues debt to fund all transfers, $B_t = (1 + r_{t-1}) B_{t-1} + T_t^h + T_t^f$

Implications of quadratic adjustment costs

- Q theory equations:

$$\frac{I_t}{K_{t-1}} - \delta = \frac{1}{\phi}\left(Q_t - 1\right)$$

$$p_t = Q_t K_t = \frac{p_{t+1} + D_{t+1}}{1 + r_t}$$

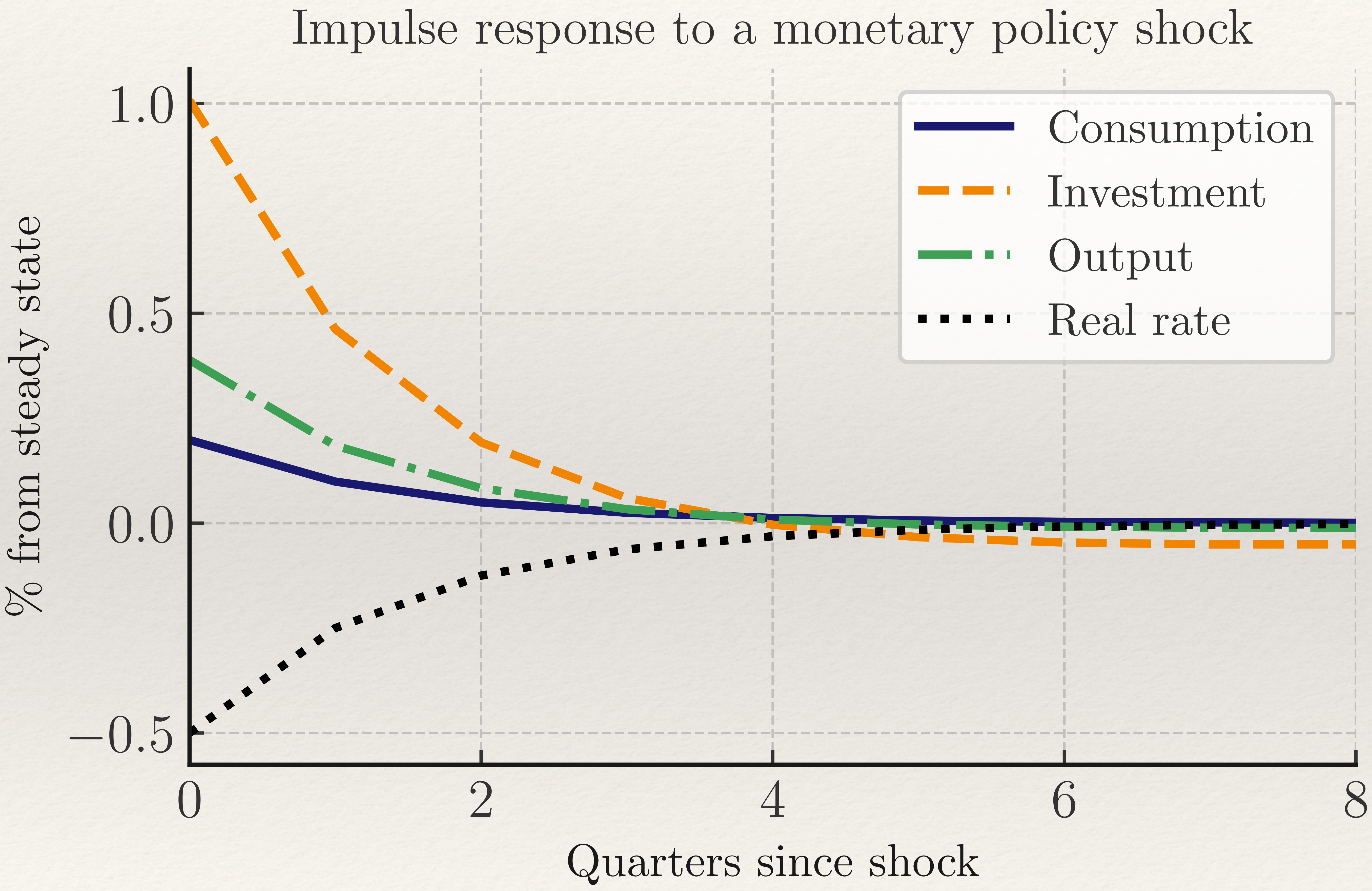
- Use mutual fund trick, assume $B = 0$ in steady state: 100% s.s. stock allocation
  - So connection to household block is $1 + r_0^p = \dfrac{p_0 + d_0}{p_{ss}}$
- Asset market clearing $A_t = p_t + B_t$. Goods market clearing: $C_t + I_t = Y_t = K_{t-1}^{\alpha} N_t^{\nu}$

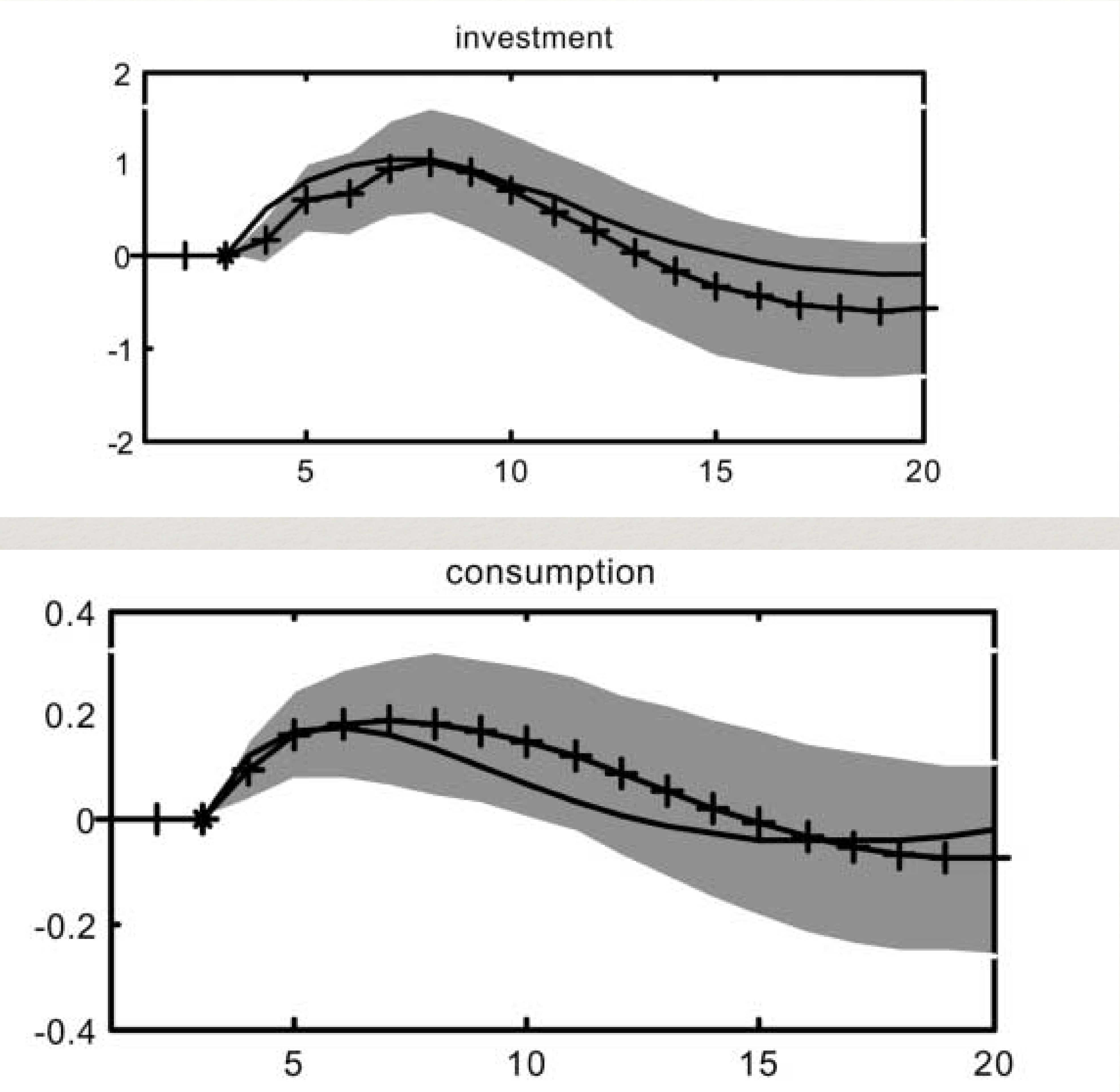
Calibration of HANK model with investment

- Standard functional forms: $u(C) = \dfrac{C^{1-\sigma}}{1-\sigma}$, adj cost for firms $\dfrac{1}{2}\phi x^2$
- Set $\alpha, \nu, \delta, \psi$ exogenously (here $\alpha + \nu < 1$), β to hit $r = 1\,\%$ quarterly
- Pick remaining 2 parameters to hit the **peak** impulse response to an identified monetary policy shock [in spirit of Christiano-Eichenbaum-Evans]
  - Elasticity of intertemporal substitution $1/\sigma$ controls consumption response
  - Degree of capital adjustment costs ϕ controls investment response

α	ν	δ	ψ	β	σ	ϕ
0.32	0.6	2.5%	1	0.99	1.6	9

Impulse response to monetary policy shock





I accounts for ~60% of the *Y* response, *C* for ~40%

source: Christiano, Eichenbaum, Evans 2005

Transmission mechanism to I

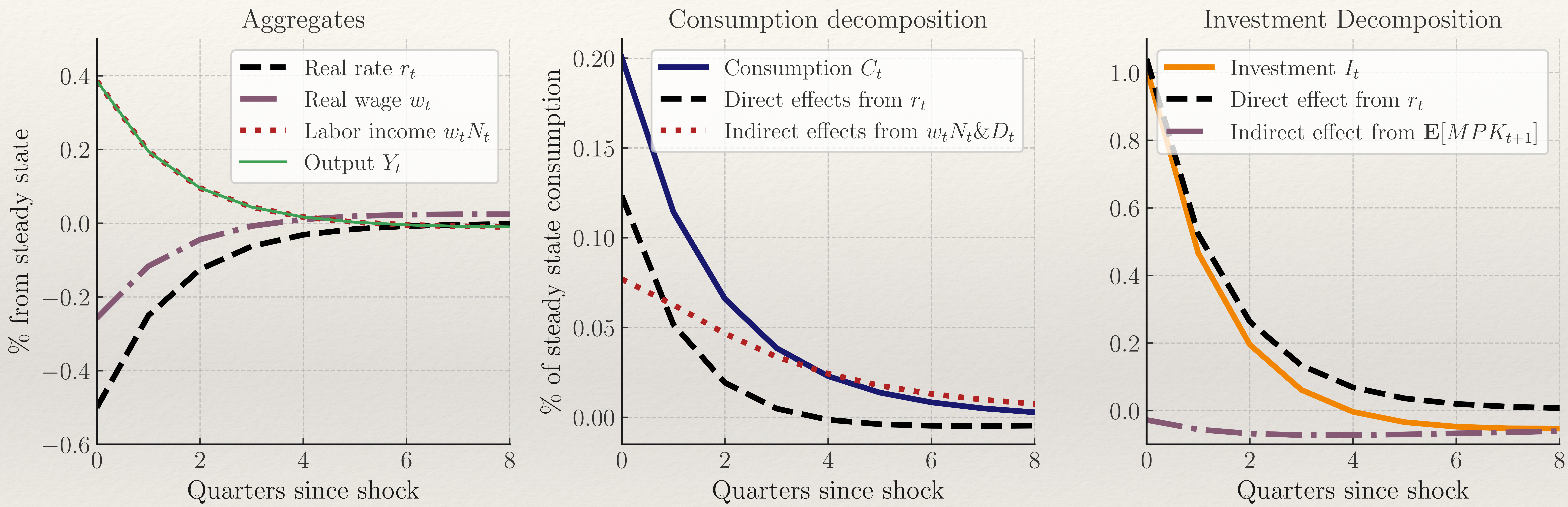
- Using Q eqns, get decomposition of investment response similar to usual for C:

$$\frac{I_t}{K_{t-1}} - \delta = \equiv \frac{1}{\phi}\left(\frac{1}{1+r_t}\left(MPK(w_{t+1}, K_t) + 1 - \delta - \frac{\phi}{2}\left(\frac{I_{t+1}}{K_t} - \delta\right)^2 + \phi\left(\frac{I_{t+1}}{K_t} - \delta\right)\frac{K_{t+1}}{K_t}\right) - 1\right)$$

$\rightarrow$ discounting channel (r_t^*) + expected MPK channel (w_{t+1})

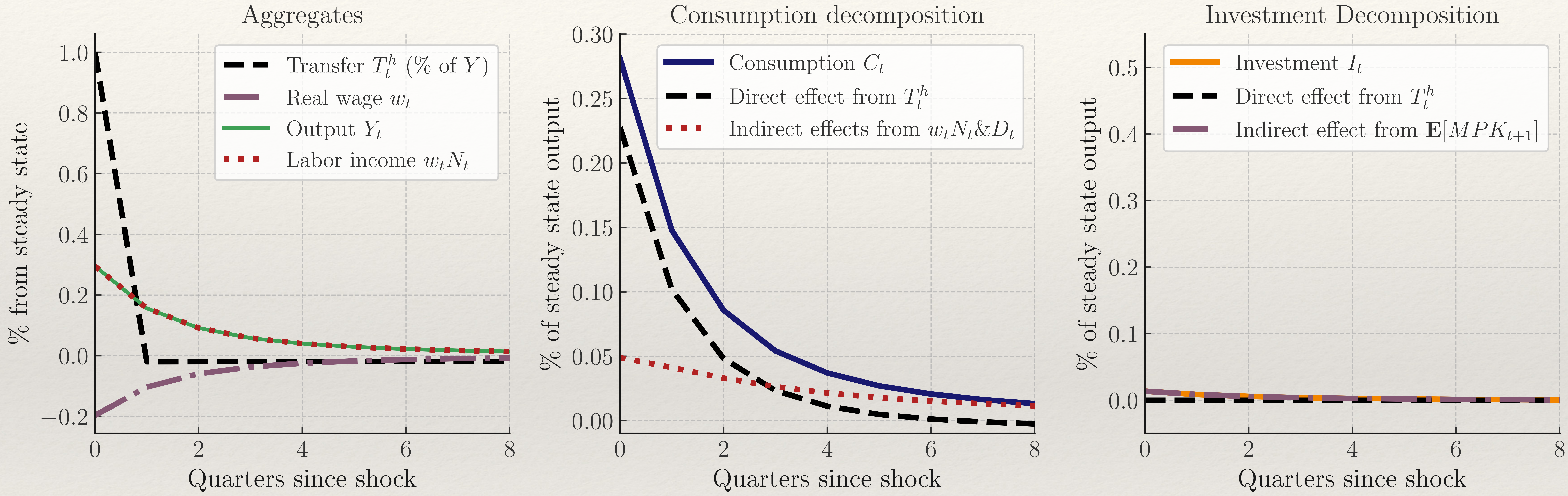
Magnitudes governed by (inverse) adjustment cost $1/\phi$

Impulse response to monetary shock



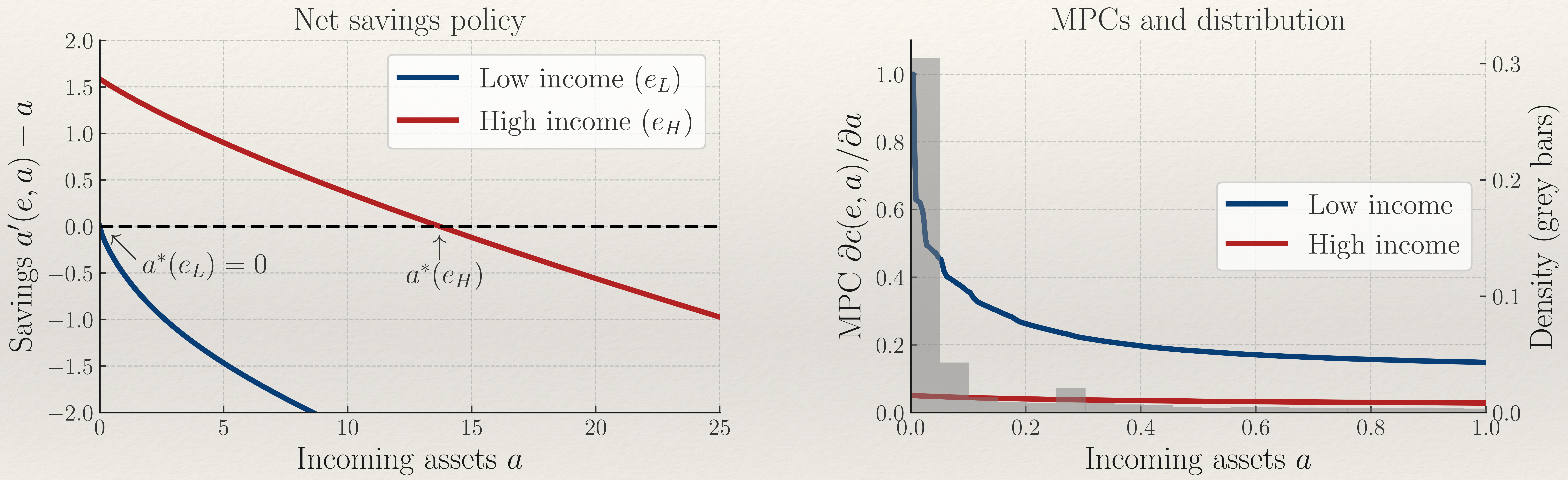
- Investment is exclusively driven by direct effects! (Here indirect effect < 0)
- Similar to representative household consumption.

Impulse response to household transfer shock



- Fiscal transfers to hh have large & persistent effects on C… but no effect on I
- Fiscal transfers to firms have **no effect** because inside = outside liquidity (MPI=0)
- **Next**: what does firm heterogeneity do?

Under hh. hood: policies, MPCs, and distribution



- Buffer-stock behavior generates stationary distribution over assets
- Concave consumption function: MPCs negatively correlated with cash-on-hand

Adding heterogenous firms

Heterogeneous firm model

- To get higher *MPI*s, build on tradition of models with idiosyncratic productivity risk + entry-exit + financial frictions [Hopenhayn; Khan-Thomas; Ottonello-Winberry…]
- Firm in idiosyncratic productivity state z with incoming capital k and debt b solves

$$v_t(z,k,b) = \max_{n,k',b',d} d + \frac{1}{1+r_t}\mathbb{E}_t\left[\theta v_{t+1}(z',k',b') + (1-\theta)\, v_{t+1}^{end}(z',k',b')\right]$$

Exit prevents firms from growing out of constraints

$$d = zk^{\alpha}n^{\nu} - w_t n - \left(k' - (1-\delta)k\right) - \varphi\left(\frac{k'-k}{k}\right)k - b + \frac{b'}{1+r_t} + T_t^f$$

Cannot issue equity
Multiplier λ on equity issuance

$$d \geq 0,\ [\lambda] \qquad b' \leq \chi k'\ [\mu]$$

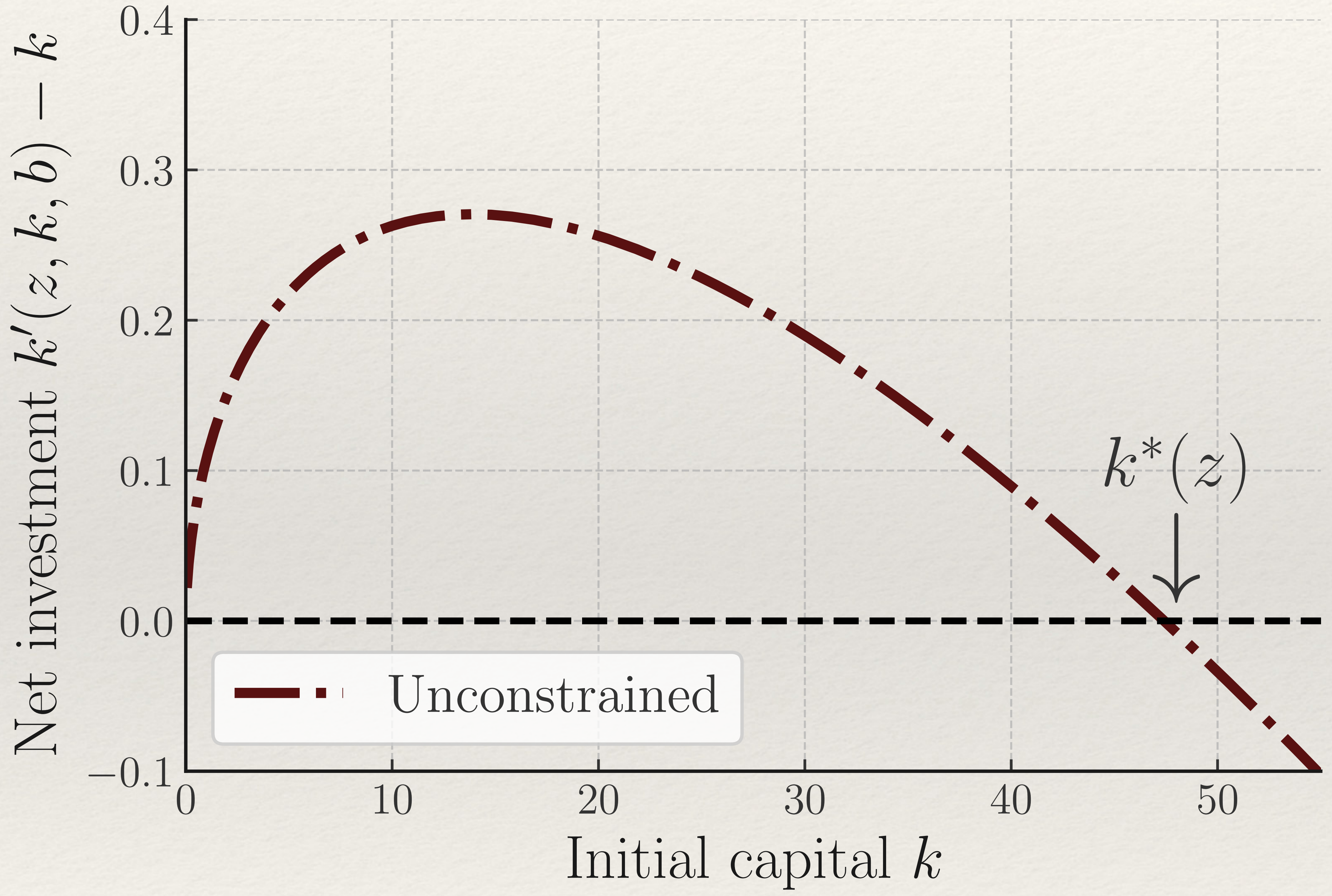
Can borrow in addition to retaining earnings subject to classic collateral constraint

$$v_t^{end}(z,k,b) = zk^{\alpha}n^{\nu} - w_t n + (1-\delta)k - b + T_t^f$$

Calibration

- Additional parameters relative to representative firm model
  - Death probability $1 - \theta = 2\,\%$ ← firm exit rate [Business Dynamics Statistics]
  - Firm productivity process ← estimated log AR(1) [Syverson 2011]
  - $\chi = 0.35$ ← ability to recover assets in bankruptcy [Kermani-Ma 2023]
  - Initial conditions $(k_0, b_0 = \chi k_0)$ ← size of new entrants << average firm
- Calibration generates average *MPI* = 0.50; ideally could use this as a data moment
- Not readily available: corporate finance *MPI* literature hasn't yet given us reliable estimates for broad set of firms (contrast with household finance *MPC* literature!)
  - [Some promising work using surveys or semi-structural estimation]

Steady state capital policy



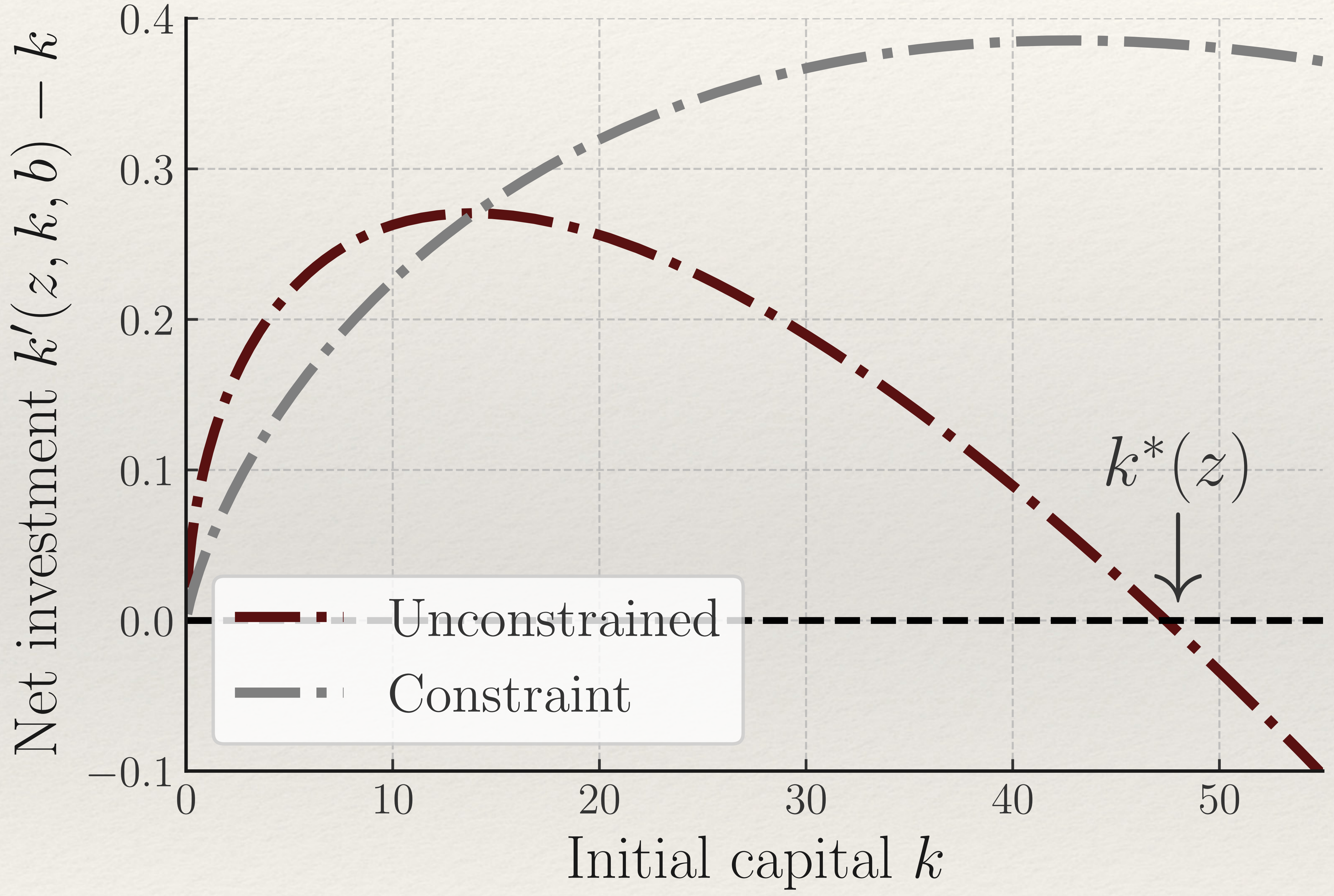
FOC for unconstrained (assuming no exit, $\theta = 1$):

$$\frac{k'-k}{k} = \frac{1}{\phi}\left(\frac{1}{1+r_t}\mathbb{E}\left[MPK(z',k',w_{t+1}) + \tilde{\varphi}(k',k'')\right] - 1\right)$$

Like rep. firm, unconstrained affected by m.p. via discounting and expected MPK effects

- Consider an unconstrained firm ($b \ll 0$)
- Diminishing returns + adjustment costs generate target $k^*(z)$

Steady state capital policy



For borrowing and dividend constrained firm:

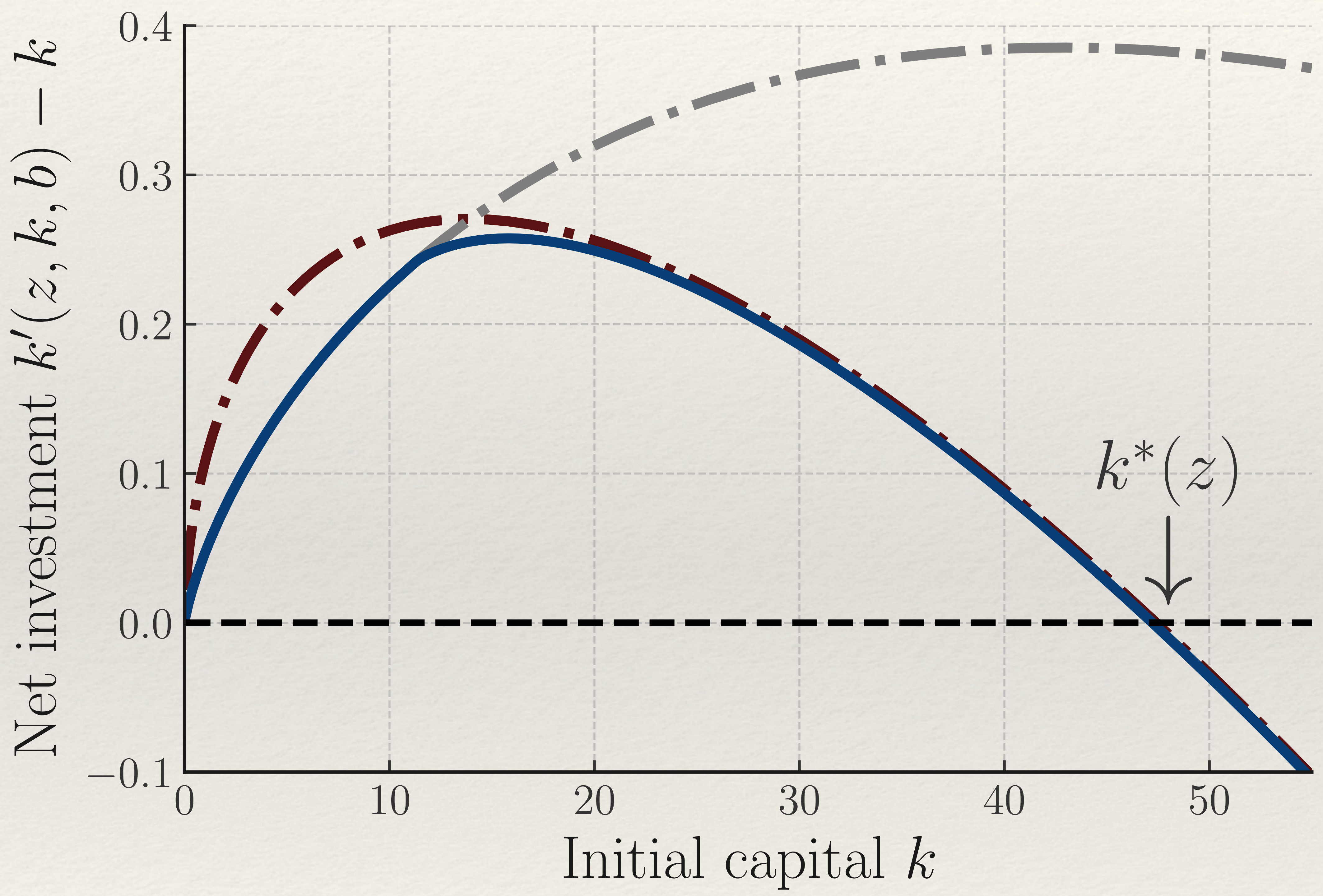
$$k' - (1-\delta)k + \varphi\left(\frac{k'-k}{k}\right)k = \pi_t(z,k) + T_t^f - b + \frac{\chi k'}{1+r_t}$$

where π_t is profits, so $k' = \bar{k}\left(z, k, b; r_t, w_t, T_t^f\right)$

Constrained firms affected by a cash flow effect: lower w_t today → higher π_t → higher I_t

- Next consider a firm that pays $d = 0$ and borrows maximally $b' = \chi k'$

Steady state capital policy



Multiplier dynamics (with $\theta = 1$)

$$\lambda_t(z,k,b) = (1+r_t)\mu_t(z,k,b) + \mathbb{E}[\lambda_{t+1}(z',k',b')]$$

pay divs only if can avoid constraint forever

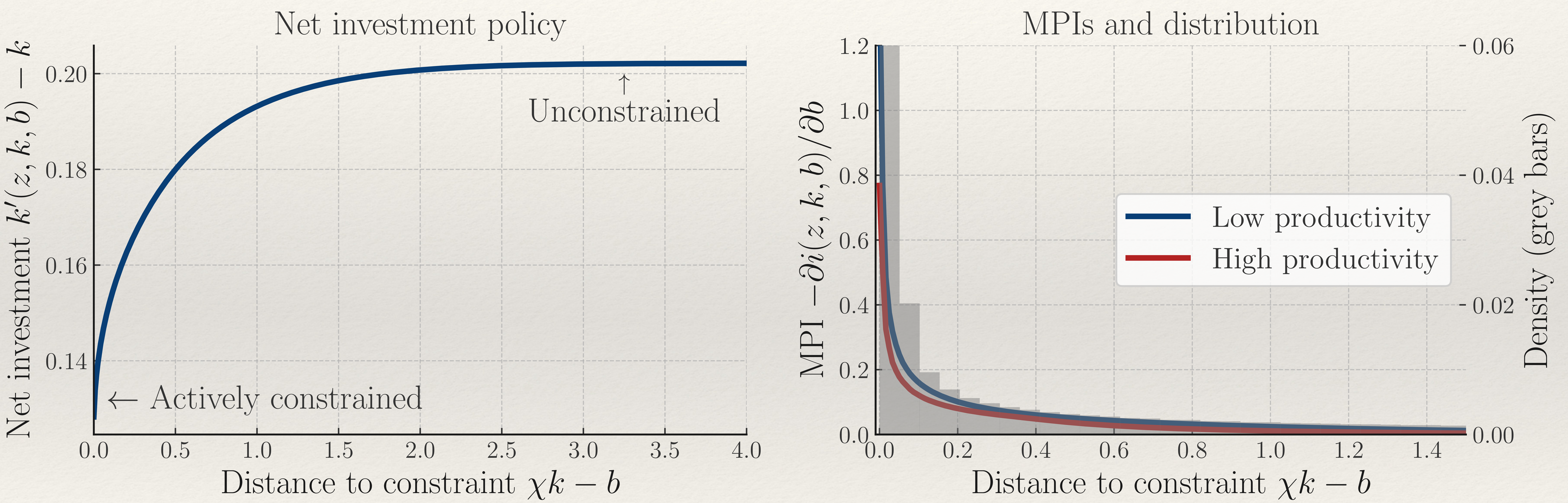
General investment FOC:

$$\frac{k'-k}{k} = \frac{1}{\phi}\frac{1}{1+r_t}\mathbb{E}\left[\frac{\lambda_{t+1}(z',k',b')}{\lambda_t(z,k,b)}\left(MPK(z',k',w_{t+1}) + \tilde{\varphi}(k',k'')\right)\right]$$

Variation in λ acts like effective risk aversion

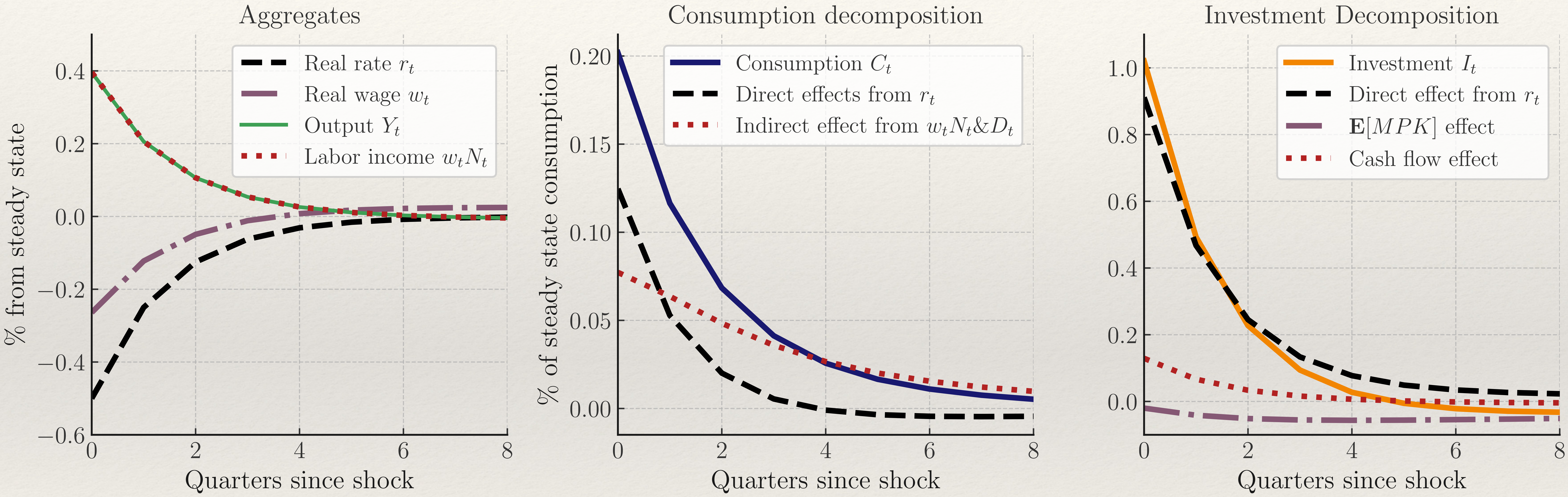
- Full policy pastes the two, with additional intermediate $\lambda > 0$, $b' < \chi k'$ region

Investment policy, MPIs and distribution



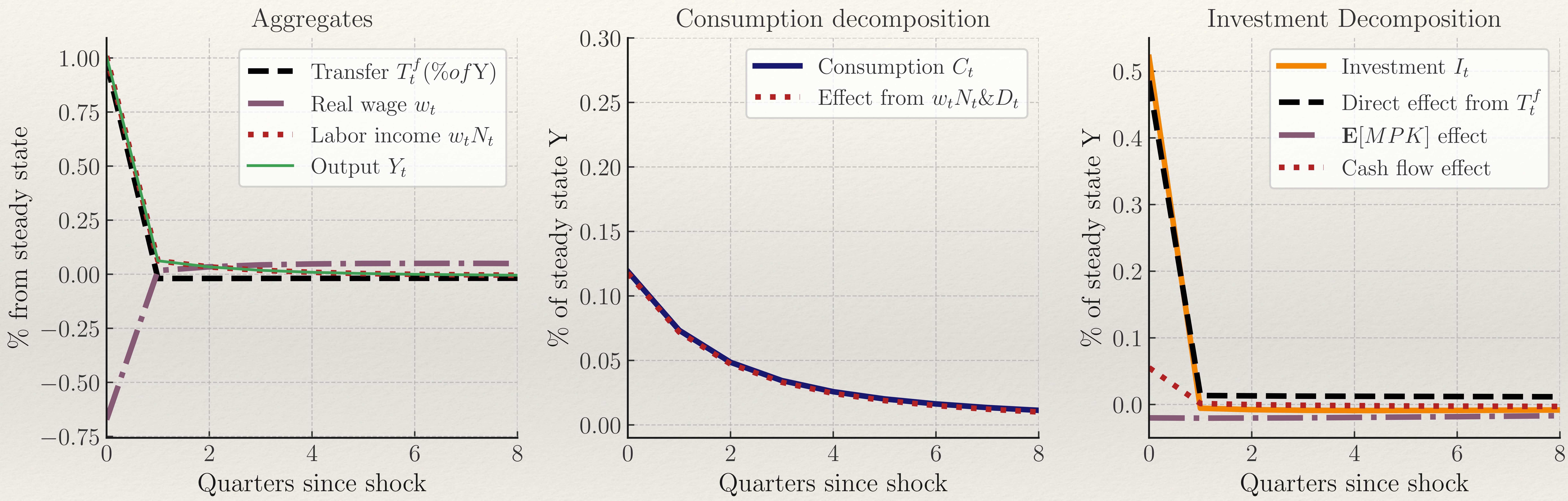
- Concave investment function: MPIs negatively correlated with dist. to constraint
- *MPI* > 1 possible due to ability to lever up $1/(1-\chi)$ - contrary to household model

Impulse response to monetary shock



- New cash flow channel of monetary policy to investment: $w_t \downarrow \rightarrow \pi_t \uparrow \rightarrow I_t \uparrow$
- Magnitude controlled by profit-weighted *MPI* (since $\pi_{it} \propto Y_t$). Here, $\overline{MPI}$ small.

Impulse response to firm transfer shock



- Inside ≠ outside liquidity: PV-0 transfers to firms boost output today!
- Large direct effect from firm transfer (large *MPI*), small multiplier (small $\overline{MPI}$)

Analogies…

	Heterogeneous households	Heterogeneous firms
Steady state	MPCs / iMPCs	MPIs / iMPIs
	Concave consumption function out of liquidity	Concave investment function out of free cash flow
	Prudence	Concern about hitting constraints
	Target (buffer) stock of assets	Target stock of capital (diminishing returns)
	Impatience	Life cycle (entry-exit)
Response to shocks	Income vs substitution effects	Income vs substitution effects
	Indirect effects of monetary policy	Cash flow effects of monetary policy
	Departure from Ricardian equivalence	Departure from inside = outside liquidity
	Income-weighted MPC governs multiplier	Profit-weighted MPI governs multiplier

How much do we get from the combination?

Households vs firm transfers

Impact on GDP (net of adj. cost) of 1% of GDP transfer…

To households

	Rep firm	Het firm
Rep household	0	0
Het household	0.29	0.31
	$\frac{MPC}{1-\overline{MPC}}$	$\frac{MPC}{1-(\overline{MPC}+\overline{MPI})}$

To firms

	Rep firm	Het firm
Rep household	0	0.52 $\frac{MPI}{1-\overline{MPI}}$
Het household	0	0.64
		$\frac{MPI}{1-(\overline{MPC}+\overline{MPI})}$

- Implications for stimulus policy ultimately depend on our views of *MPC*s vs *MPI*s !

Bringing firm heterogeneity to HANK

- Firms have been left out but have their rightful place in HANK !
- With non-zero *MPI*s as in the data:
  - … cash flow transmission channel to I, paralleling indirect effect on C
  - … PV-0 transfers to firms have large and persistent effects on Y
  - … activates Keynesian feedback effects: $Y \rightarrow I \rightarrow Y \rightarrow C$ and $Y \rightarrow C \rightarrow Y \rightarrow I$
- More empirical work needed to measure MPIs & covariance with firm observables